

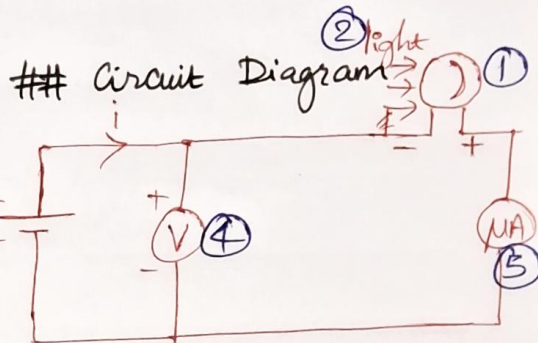
Planck's constant

Aim

To determine Planck's constant and work function of a photo sensitive material using photo-electric effect.

Apparatus

- * ① Vacuum phototube
- ② Light source
- ③ Colour filters
- ④ Power supply
- ⑤ Digital micro ammeter

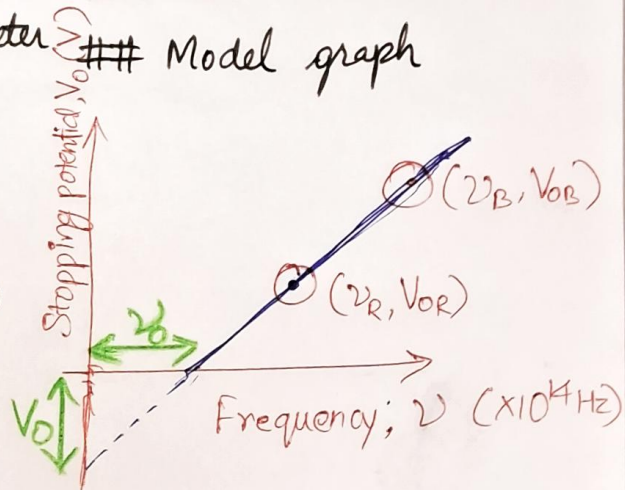


Formula

①: $h = \frac{e(V_{0B} - V_{0R})}{\nu_B - \nu_R}$

②: $W = h \times (\text{x-intercept})$

Quantity	Symbol	Unit
Planck's constant	h	Js
Charge of electron	e	$1.6 \times 10^{-19} \text{ C}$
Stopping potential	V_0	V
Stopping pot. for blue light	V_{0B}	V
Stopping pot. for red light	V_{0R}	V
Frequency of blue	ν_B	Hz
Frequency of red	ν_R	Hz



Tabular form

S.No.	Color	Wavelength (nm)	Frequency ($\times 10^{14} \text{ Hz}$)	Stopping potential (V)
1				
2				
⋮				

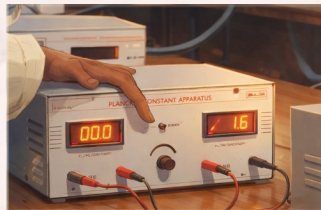
③

Threshold frequency	ν_0	Hz
or		
x intercept		
Work function	W	eV
or		
y intercept		

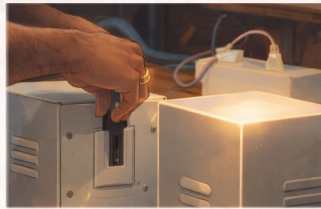
Procedure

- ① @LIGHT OFF condition,
zero the current.
- ② Place the filter.
- ③a) Switch on the light source
and @LIGHT ON condition,
- ③b) zero the current.
- ④ Note the stopping voltage.
- ⑤ Change filter and goto 3b.

1



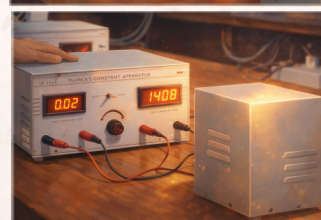
2



3a



3b



Precautions

1. Be aware of connecting correct polarity of components.
2. Light source should illuminate photocell maximally.

Calculations

Result

1. The work function of photocell material is _____ eV.
2. The Planck's constant is found to be _____ Js.

Applications

- Photoelectric effect is applicable in photo detection.
like charge coupled devices (CCDs). These are used in digital cameras
- Radioactivity detection by conversion of ionizing radiation into electric current

Viva (External)

- What is stopping potential?
- Describe photo-electric effect